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Video Shot Classification Using 2D Motion Histogram

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***Abstract*—This paper describes a new histogram based approach for classifying camera motions or characterizing video shots. The proposed method utilizes both magnitudes and orientations of motion vectors simultaneously, rather than using them separately as same as the existing methods. A 2D histogram, namely 2D array motion vector histogram, that carries both magnitudes and orientations of motion vectors detected in video sequences. We have compared the proposed approach with existing methods. Experimental results show that the proposed method can classify more camera motions with better performance.**

Keywords-Camera motion estimation; histogram; shot classification;

I. Introduction

Shot classification is an approach in a trend of computer visual. A motion vector field (MVF) basically has four camera motion patterns: (1) stationary, (2) panning, (3) tilting, and (4) zooming. If the MVF does not follow one of the patterns, it is known as a scene change. It happens when a sudden change of scenes or switching in camera. More than a decade year, researchers had tried to characterize the camera motions in MPEG compressed videos directly [1]. The scene changes are detected by comparing two histograms of I frames in intensity and motion vectors (MV). The MVFs on B and P frames are used to describe the camera motions. Although it achieves in computational time, using MVs from the compressed videos can be misleading in results. Recently, there are accomplished works in shot characterization based on MV histogram. In the existing work [2], it removes foreground MVs from the MVF because it is not reliable to the camera motions. Then the MVF is divided into nine sub-regions equally to find the camera motion patterns. Although it performs high accuracy, it cost highly computational time due to iterative calculating the histograms of each sub-region. In existing work [3], two MV histograms, magnitude and orientation, are used to detect the camera motions. However, it cannot identify the camera motions if the video frame is covered with the large foreground.

This paper purposes a method to identify camera motions from the MVFs. A 2D array MV histogram is presented in this paper. It is built by merging two MV histograms of magnitude and orientation. Experimental results show that the proposed method can classify more camera motions comparing to the existing methods.

The rest of the paper is organized as follows. Section II describes the proposed method for camera motion estimation. Section III shows the experimental results of our proposed method. Finally, the conclusions are described in Section IV

II. Proposed Method

Since we focus on the algorithm that can describe the camera motions from the MVFs based on the histogram. To achieve our goal, we use MATLAB to conduct experiments. The proposed method has three main steps. First, a MVF is estimated from video's current frame and its previous adjacent frame. Second, the 2D array MV histogram is constructed the MVF. Finally, the 2D array MV histogram is used to describe camera motions.

A. Motion vector field estimation

The adaptive rood pattern search (ARPS), fast block-based motion estimation, is used since it can estimate both large motion and small MV in video sequences. It gives performance better than existing fast block-based motion estimations which is described in [4]. Recently, the ARPS is applied to a large-scale implementation that describe in [5]. The ARPS's MATLAB source code is available on the website [6].

B. 2D array motion vector histogram

Fig. 1 shows an example of the 2D array motion vector histogram which its size is 7 by 7. Each block contains MV information (m, θ) that is calculated by Eqs. (1) and (2) where m is the magnitude and θ is the orientation of the MV (u, v).

$$m = \sqrt{u^2 + v^2} \tag{1}$$

$$\theta = \arctan\left(\frac{v}{u}\right) \tag{2}$$

(4, 135)	(4, 123)	(3, 108)	(3, 90)	(3, 71)	(4, 56)	(4, 45)
(4, 146)	(3, 135)	(2, 116)	(2, 90)	(2, 63)	(3, 45)	(4, 33)
(3, 161)	(2, 153)	(1, 135)	(1, 90)	(1, 45)	(2, 26)	(3, 18)
(3, 180)	(2, 180)	(1, 180)	(0, 0)	(1, 0)	(2, 0)	(3, 0)
(3, 198)	(2, 206)	(1, 225)	(1, 270)	(1, 315)	(2, 333)	(3, 341)
(4, 213)	(3, 225)	(2, 243)	(2, 270)	(2, 296)	(3, 315)	(4, 326)
(4, 225)	(4, 236)	(3, 251)	(3, 270)	(3, 288)	(4, 303)	(4, 315)

Fig. 1. An example of 2D array motion vector histogram size 7×7.

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C. Camera motion estimation

First of all, the orientations are divided into 8 groups as shown in Fig. 2 since the ARPS provides opposite MVs direction to the camera motions. Consequently, the values of the 2D array motion vector histogram are normalized in range between 0 and 1. If the normalized values are less than 0.14, we ignore them. This threshold value is selected through the experiments of the project. The thresholded histogram is visualized by black color the ignored block as while white color is the interested block.

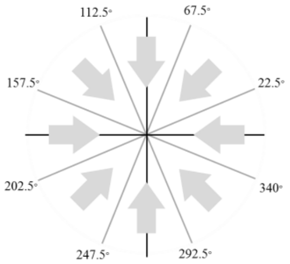

Fig. 2. Camera motion directions for classifying.

We focus on the interested regions (i.e. white regions) in the thresholded histogram in order to classify the camera motions. There are three cases depending on their profiles. We briefly explain them via the decision tree in Fig. 3.

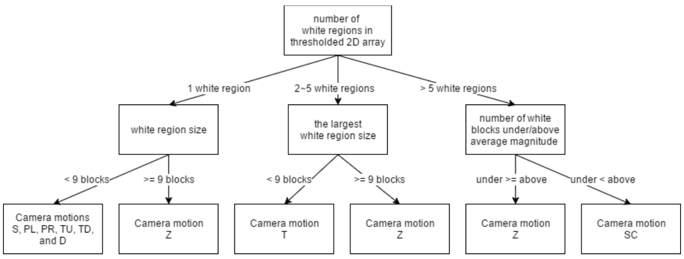

Fig. 3. Decision tree of the camera motion estimation.

First case, there is one white region. If its size is smaller than 9 blocks, we check its location using magnitude m. If the magnitude m is zero (i.e. center), we conclude that is the stationary. If the magnitude m is non-zero (i.e. off-center), we consider as the panning or tilting camera motions belonging to its orientation θ. If its size is greater than or equal to 9 blocks, it considers as the zooming camera motion.

Second case, there are two to five white regions. The zooming camera motion is verified by checking the largest region's size. If it is greater than or equal to 9 blocks, the proposed method concludes that is zooming camera motion. Otherwise, is tracking camera motion.

Third case, the number of the white regions is greater than five regions. The average magnitude $\bar{m}$ is calculated in order to classify between zooming camera motion and tracking camera motion. The white blocks are separated into two groups: first, the white blocks that has magnitude m less than or equal to the average magnitude m. Second, first, the white blocks that has magnitude m greater than the average magnitude m. If the number of white blocks in the first group is greater than or equal to the second group, we conclude that is zooming camera motion. Otherwise, it is scene change camera motion.

To distinguish between zooming in and zooming out, the MVF is divided into four sub-regions. In each sub-region, the MV's orientation calculated by the ARPS is compared with the MV's orientation in template following in Fig.4. If it matches at least 50%, it votes for zooming in for the template in Fig. 4(a) and votes for zooming out for the template in Fig. 4(b).

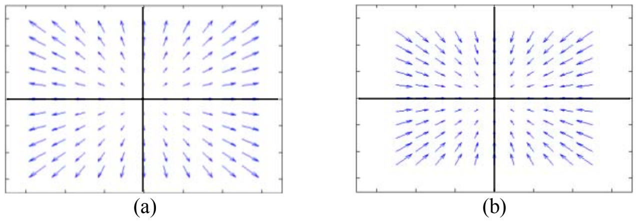

Fig. 4. Motion vector field patterns (a) zooming in (b) zooming out.

III. RESULTS AND DISCUSSION

We use video dataset from websites [7] [8]. We apply the ARPS with a suitable square window search in order to retrieve an appropriate MVF from the video dataset.

A. Experiments

Fig. 5 shows an example of a scene change. The video frame suddenly changes from Fig. 5(a) to Fig. 5(b). Fig. 5(c) shows the non-uniform MVF between these two frames. Fig. 5(d) shows the temperature distribution of the 2D array histogram where zero represents blue color while one represents red color. From the temperature distribution, array blocks, which are yellow and red colors, separate away from each other because the MV information of each array block is completely different. Fig. 5(e) illustrates the thresholded histogram that a red dot is located at the center of the array. The proposed method considers as the scene change by referring to the third step as mention the previous section.

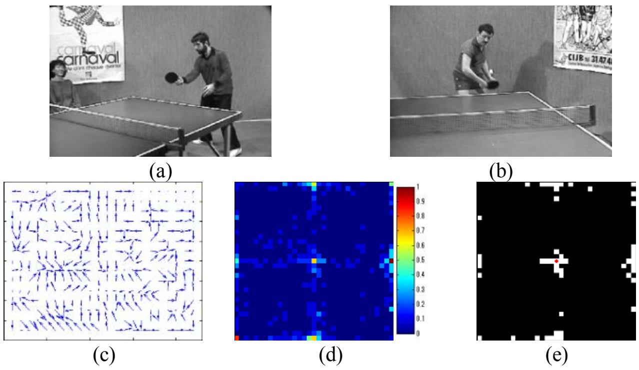

Fig. 5. Camera motion estimation of scene change. (a) reference frame, (b) current frame, (c) motion vector field, (d) 2D array motion vector histogram in temperature distribution, and (e) thresholded histogram.

Fig. 6 shows an example of zooming camera motion. From the reference frame in Fig. 6(a), the camera adjusts focal length to zoom out that transforms to Fig. 6(b). Fig. 6(c) is the MVF between these two frames which MVs diverge away from the center of the MVF. By the MVF, the temperature distribution has a rectangle shape of heat blocks that has each MV (m, θ) similarly as shown in Fig. 6(d). There is a white region after thresholding following in Fig. 6(e). Thus, the proposed method consider as zooming camera motion refer to the first step in the previous section.

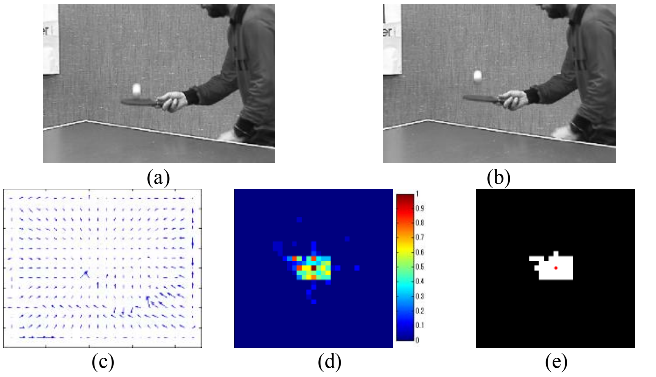

Fig. 6. Camera motion estimation of zooming. (a) reference frame, (b) current frame, (c) motion vector field, (d) 2D array motion vector histogram in temperature distribution, and (e) thresholded histogram.

Fig. 7 shows an example of tracking camera motion. The video frame in Fig. 7(a) is transformed to Fig. 7(b) by following the helicopter in upper-left direction. On the MVF in Fig. 7(c), there are two kinds of the MV: first, non-stationary MV that is on the background. Second, nearly stationary MV that is on the foreground. Fig. 7(d) shows two heat spots on the temperature distribution which one heat spot is on the off-center while the another spot is on the center. Fig. 7(e) illustrates two white regions on the thresholded histogram. The proposed method concludes as the tracking camera motion following to the second case in the previous section.

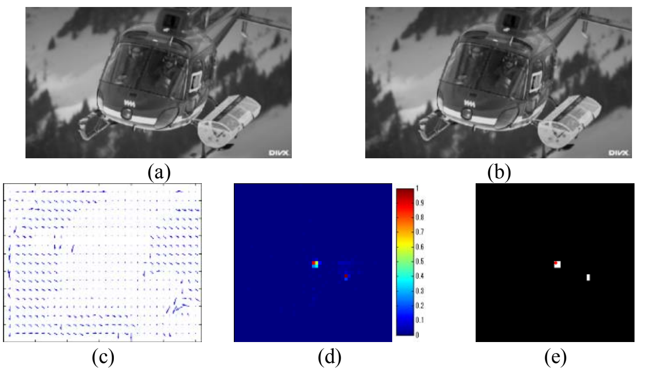

Fig. 7. Camera motion estimation of tracking. (a) reference frame, (b) current frame, (c) motion vector field, (d) 2D array motion vector histogram in temperature distribution, and (e) thresholded histogram.

B. *Comparisons with existing methods*

There are experiment data and video dataset by Duan's method and Okade's method [9] [10]. The two video sequences namely, stefan and desert, are used for comparison. There are changing in ground truth on the previous version. In the video sequence stefan, between frames 179 and 185, and between frames 231 and 269, from our observation, they are combination between zooming and panning camera motions. Following Figs. 8 and 9, they are obvious that the backgrounds are shifted to left and right while their size is slightly changed. Thus, these ground truths are changed from the zooming camera motion to the panning camera motion.

Fig. 8. Stefan video sequence between frame numbers 179 and 184.

Fig. 9. Stefan video sequence between frame numbers 232 and 237.

Table I is a comparison in camera motion between the proposed method and the existing methods using the two video sequences. The abbreviation in each table cell is given by stationary (S), panning right (PR) panning left (PL), tilting up (TU), tilting down (TD), diagonal panning (D), zooming (Z), tracking (T), accuracy (ACC), sensitivity (SEN), precision (PRE), specificity (SPE), and F1 score (F1). From the comparison, it is obvious that the existing methods mistake the non-S camera motions to be the S camera motion. The existing methods cannot detect the camera motion in diagonal direction while the proposed method can do it. Therefore, their F1 scores in the D camera motion are zero and their F1 score in the TU, and TD are low. In overall, the performance is ranked by the proposed method > Okade > Duan.

TABLE I PERFORMANCE COMPARISON IN VIDEO SEQUENCES STEFAN AND DESERT

Ground truth	Duan's method [10]							Okade's method [3]							Proposed method							
	S	PR	TU	PL	TD	D	Z	S	PR	TU	PL	TD	D	Z	S	PR	TU	PL	TD	D	Z	T
S	41	5	0	5	0	0	4	53	1	0	0	0	0	1	48	2	0	2	1	1	0	0
PR	2	111	1	2	0	0	5	3	111	0	0	0	0	7	2	120	0	0	0	0	0	2
TU	5	0	11	0	0	0	0	1	0	15	0	0	0	0	0	0	16	0	0	0	0	0
PL	11	1	0	175	0	0	66	2	1	3	209	0	0	38	2	0	0	247	0	3	0	2
TD	0	0	0	0	10	0	6	3	0	0	0	13	0	0	0	0	0	0	16	0	0	0
D	2	1	19	3	21	0	18	2	6	18	10	28	0	0	0	1	0	6	0	57	0	0
Z	0	0	0	2	0	0	9	0	0	0	0	0	0	11	0	0	0	0	0	0	11	0
ACC	93.7	96.8	95.3	83.2	95.0	88.1	81.2	97.6	96.6	95.9	89.9	94.2	88.1	91.4	98.1	99.1	100	97.2	99.8	98.0	100	-
SEN	74.5	91.7	68.8	69.2	62.5	0.0	81.8	96.4	91.7	93.8	82.6	81.3	0.0	100	88.9	98.4	100	97.2	100	89.1	100	-
PRE	67.2	94.1	35.5	93.6	32.3	-	8.3	82.8	93.3	41.7	95.4	31.7	-	19.3	92.3	97.6	100	96.9	94.1	93.4	100	-
SPE	95.8	98.3	96.2	95.8	96.0	100	81.1	97.7	98.1	96.0	96.5	94.6	100	91.2	99.2	99.3	100	97.2	99.8	99.2	100	-
F1	70.7	92.9	46.8	79.5	42.6	0.0	15.1	89.1	92.5	57.7	88.6	45.6	0.0	32.4	90.6	98.0	100	97.1	97.0	91.2	100	-

C. *Quantitative evaluation*

Table II shows the statistic of several camera motions from the video dataset. There are totally 14,453 frames which are from several categories. There are 201 frames of scene changes. The S, PL, and PR camera motions have large numbers since they are the most general camera motions while TU and TD camera motions have low numbers since they are mostly available in the sports videos. For zooming in (ZI), zooming out (ZO) and T, they have low numbers since they are used for specific situations. Table III summarizes the results in ACC, SEN, PRE, SPE, and F1.

The proposed method detects the S camera motion wrongly about 1%. From 3,082 frames of PR, PL, TU, TD, and D camera motions, the proposed method can detect 2,845 frames correctly. Most false results come from the video frames that are covered by a large foreground. Thus, there are less background motion vectors for classifying these camera motions. For ZI and ZO camera motions, the proposed method can detect them for 321 frames from 350 frames without mistaking in each other. The 29 false results mostly come from very slow in speed of ZI or ZO camera motions. The missing 130 frames of T camera motion are caused by the tracking objects move in direction and speed similar to the camera. Moreover, there is no background in the video frames since they are covered by the foreground. Thus, they are easily to consider as the SC camera motion. In SC camera motions, there are two kinds of them, suddenly SC and animation SC. There is no error in results of suddenly SC since the ARPS cannot find the similar image block during searching while the ARPS can find the similar image block for animation SC. Thus, the animation SC may mislead to the uniform MVF patterns rather than the non-uniform MVF patterns.

TABLE II STATISTICS OF DIFFERENT CAMERA MOTIONS OF 14,453 FRAMES FROM THE VIDEO DATASET

	Camera motions									
	S	PR	TU	PL	TD	D	ZI	ZO	T	SC
Total	10379	1078	445	1137	281	141	183	167	441	201

TABLE III CLASSIFICATION RESULTS OF 10 CAMERA MOTIONS

Ground truth	Proposed method									
	S	PR	TU	PL	TD	D	ZI	ZO	T	SC
S	10195	16	18	44	31	4	8	3	34	26
PR	2	1072	0	0	0	4	0	0	0	0
TU	0	1	322	0	0	4	30	15	53	20
PL	4	0	0	1093	0	3	0	0	31	6
TD	35	0	2	0	228	0	2	2	8	4
D	0	2	2	6	1	130	0	0	0	0
ZI	14	2	0	2	0	0	164	0	1	0
ZO	3	1	0	0	0	0	0	157	0	6
T	10	54	4	3	9	2	16	4	311	28
SC	25	7	1	0	0	0	4	0	7	157
ACC	98.1	99.4	99.0	99.3	99.3	99.8	99.5	99.8	98.2	99.1
SEN	98.2	99.4	72.4	96.1	81.1	92.2	89.6	94.0	70.5	78.1
PRE	99.1	92.8	92.3	95.2	84.8	88.4	73.2	86.7	69.9	63.6
SPE	97.7	99.4	99.8	99.6	99.7	99.9	99.6	99.8	99.0	99.4
F1	98.7	96.0	81.1	95.7	82.9	90.3	80.6	90.2	70.2	70.1

IV. CONCLUSION

In this paper, a 2D array motion vector histogram is presented to classify the camera motions for video sequences. The camera motions are described by using thresholded histogram to analyze the MVF via the number of interested regions (i.e. white regions) and its size. For example, MVs are detected in a converging or diverging manner across the entire video frame when the camera motion is ZI or ZO. Then the MVs produce a squared shape region in the 2D array motion vector histogram because their magnitudes are similar to each other while their orientations cover all directions. The proposed method can distinguish two zooming camera operations, ZI (i.e. diverging MVF) and ZO (i.e. converging MVF), which consumes less computational time than the existing work [2] because S, PR, PL, TU, TD, D, T, and SC camera motions have no need to compare their MV orientation iteratively. As future work, we plan to investigate multiple camera motions (e.g. ZI with PR).

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